eLife

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted
August 21, 2023 (this version)

Posted to bioRxiv
July 22, 2023

Sent for peer review
July 5, 2023

Neuroscience

Serotonergic and dopaminergic neurons in the dorsal raphe are differentially altered in a mouse model for Parkinson's disease

Laura Boi, Yvonne Johansson, Raffaella Tonini, Rosario Moratalla, Gilberto Fisone ✉, Gilad Silberberg ✉

Department of Neuroscience, Karolinska Institutet, Stockholm 17177, Sweden • Sainsbury Wellcome Centre for Neural Circuits and Behaviour, University College London, London, UK • Neuromodulation of Cortical and Subcortical Circuits Laboratory, Istituto Italiano di Tecnologia, Genova, Italy • Cajal Institute, Spanish National Research Council (CSIC), Madrid, Spain • CIBERNED, Instituto de Salud Carlos III, Madrid, Spain

Abstract

Parkinson's disease (PD) is characterized by motor impairments caused by degeneration of dopamine neurons in the substantia nigra pars compacta. In addition to these symptoms, PD patients often suffer from non-motor co-morbidities including sleep and psychiatric disturbances, which are thought to depend on concomitant alterations of serotonergic and noradrenergic transmission. A primary locus of serotonergic neurons is the dorsal raphe nucleus (DRN), providing brain-wide serotonergic input. Here, we identified electrophysiological and morphological parameters to classify serotonergic and dopaminergic neurons in the murine DRN under control conditions and in a PD model, following striatal injection of the catecholamine toxin, 6-hydroxydopamine (6-OHDA). Electrical and morphological properties of both neuronal populations were altered by 6-OHDA. In serotonergic neurons, most changes were reversed when 6-OHDA was injected in combination with desipramine, a noradrenaline reuptake inhibitor, protecting the noradrenergic terminals. Our results show that the depletion of both noradrenaline and dopamine in the 6-OHDA mouse model causes changes in the DRN neural circuitry.

eLife assessment

This **important** work identifies electrophysiologically and morphologically distinct subpopulations of dorsal raphe nucleus neurons, which are in turn, differentially impacted in a toxin-based mouse model of Parkinson's disease. The inclusion of several thoughtful controls and rigorous exclusion criteria makes the presented results highly **convincing**. These findings suggest a significant interplay between catecholaminergic systems in healthy and parkinsonian conditions, as well as neuronal structure and function. Such findings provide a strong foundation for basic scientists as well as pre-clinical researchers interested in the role of dorsal raphe neurons in Parkinson's disease.

Introduction

Parkinson's Disease (PD) is a frequent neurodegenerative disorder characterized by the progressive loss of dopaminergic neurons in the nigrostriatal pathway, leading to bradykinesia, tremor, rigidity, and postural instability [1, 2]. These cardinal motor symptoms are typically addressed by administration of dopaminergic drugs or by deep brain stimulation. PD patients also experience non-motor symptoms including sleep, affective, and cognitive dysfunctions often preceding the motor disabilities [3, 4]. These comorbidities are in large part refractory to current PD treatments and are thought to be caused by neurodegenerative processes occurring in concomitance to the loss of midbrain dopaminergic neurons. However, the pathology underlying non-motor symptoms remains poorly understood.

Post-mortem studies in PD patients provided first insights into the brain areas which might be involved in the etiology of non-motor dysfunctions in PD. Besides the profound degeneration of the substantia nigra pars compacta (SNc), these studies found cell loss and reduced neurotransmitter release in other monoaminergic brain regions, including the dorsal and median raphe nuclei (DRN and MRN, respectively), and the locus coeruleus (LC) [1, 5–8]. The DRN constitutes the main source of serotonin in the brain with serotonergic cells ($DRN^{5\text{-}HT}$) accounting for 30-50% of its neurons [9]. $DRN^{5\text{-}HT}$ neurons have been implicated in numerous neuropsychiatric diseases, rendering them a potential neural substrate for non-motor symptoms in PD. In fact, several studies have shown that serotonergic markers and transmitter levels are altered in Parkinson patients as well as in non-human primate and rodent models of PD [10–17]. Notably, alterations in the serotonergic system have also been related to non-motor comorbidities in PD [18, 19]. Yet, functional investigations of $DRN^{5\text{-}HT}$ in rodent models of PD have led to conflicting results showing both increased and decreased activity in $DRN^{5\text{-}HT}$ neurons themselves as well as in their downstream targets [20–22]. Besides the serotonergic neurons, the DRN comprises other neuronal populations, including a group of dopamine neurons (DRN^{DA}), in which Lewy bodies have been found in PD patients [6]. DRN^{DA} neurons have been linked to the regulation of pain, motivational processes, incentive memory, wakefulness and sleep-wake transitions [23–27], but their ultimate behavioral significance is yet to be elucidated [28–32]. In addition, the physiology and pathophysiology of DRN^{DA} neurons in PD remains elusive. The sparsity of research on DRN^{DA} neurons is likely due to the technical challenges associated with targeting this population among the diverse cell types in the DRN and adjacent structures (e.g., retrorubral field, periaqueductal grey and LC), which often co-express signature genes, hampering their molecular identification and region-specific manipulations with cre driver lines [9, 33–36].

Recently, this issue has been addressed by Pinto and colleagues who showed that DRN^{DA} neurons are most faithfully labelled in transgenic mice in which the expression of cre is linked to the dopamine transporter (DAT-cre) [34]. Previously, the membrane properties of DRN^{DA} neurons have only been addressed in mice in which DRN^{DA} neurons were identified based on the expression of the transcription factor Pitx3 or the enzyme tyrosine hydroxylase (TH) [33]. In Pitx3-GFP mice, about 70% of GFP-positive neurons are TH- positive as shown by immunohistochemistry. Moreover, 40% of TH-positive neurons in the DRN are not labelled in these mice, suggesting that this line targets a subpopulation of DRN^{DA} neurons [33]. The widely used TH-cre reporter line has been found to show ectopic expression of cre in non-dopaminergic neurons, probably caused by a transient developmental expression of TH [37, 38]. In addition, the TH-cre line also labels noradrenergic neurons in the neighboring LC [37], which produces most of the noradrenaline (NA) in the brain and is involved in mood control, cognition, and sleep regulation [39]. The large overlap of functions ascribed to the LC and DRN is thought to result from the complex reciprocal synaptic connections between these two brain areas: notably, the LC provides noradrenergic input to the DRN [40, 41] and also receives input from DRN$^{5\text{-}HT}$ neurons [42–44].

Here, we used ex vivo whole-cell patch clamp recordings and morphological reconstructions to characterize the electrophysiological and morphological properties of DRN^{DA} and DRN$^{5\text{-}HT}$ neurons in wild type and DAT- tdTomato mice. Moreover, we studied the impact of catecholamine depletion on DRN^{DA} and DRN$^{5\text{-}HT}$ populations in the 6-OHDA toxin model of PD.

Results

DRN^{DA} and DRN$^{5\text{-}HT}$ neurons are electrophysiologically distinct cell-types

To investigate the electrophysiological and morphological profile of DRN^{DA} neurons and to compare it to DRN$^{5\text{-}HT}$ neurons, we performed whole-cell patch clamp recordings in coronal slices of adult wild type and DAT-cre mice crossed with tdTomato reporter mice (**Figure 1A**). All neurons were filled with neurobiotin and Alexa488 while recording. Alexa488 allowed us to take snapshots of recorded neurons at different time points, thus facilitating the topographical registration of recorded neurons to the post-hoc stained slices (Suppl. Figure 1). Using this approach, we obtained complete sets of electrophysiological and morphological data from 75 neurons in the DRN. Cells were identified as DRN$^{5\text{-}HT}$ or DRN^{DA} neurons based on tryptophan hydroxylase (TPH) or TH immunoreactivity, respectively (**Figure 1A-D**). None of the recorded neurons was positive for both TPH and TH (n = 0/412). During the recordings, we used a series of depolarizing and hyperpolarizing current steps and ramps that allowed us to characterize active and passive membrane properties in detail (**Figure 1E** – G). Based on the electrophysiological data, we first tested possible differences between TH-positive neurons recorded in wild-type mice and tdTomato-positive neurons recorded in DAT-tdTomato mice. We found no differences between these two groups (n = 13 TH-positive vs n = 30 tdTomato-positive neurons, Suppl. Figure 2A) and neither within the subset of tdTomato-positive neurons when comparing TH-positive to TH-negative neurons (n = 25 TH-positive vs n = 5 TH-negative neurons, Suppl. Figure 2A–E). Out of 114 tdTomato-positive neurons only one cell displayed a different electrophysiological profile than all other DRN^{DA} neurons, suggesting a false-positive rate of 0.8%. That neuron was TH-negative and displayed profoundly distinct intrinsic properties, and was therefore excluded (Suppl. Figure 2F, G). Taken together, the electrophysiological results support the use of the DAT-tdTomato mouse line when studying DRN^{DA} neurons and data from both mouse lines were pooled. Recordings of DRN^{DA} neurons showed that they share several properties characteristic of other dopaminergic populations located in the SNc and the ventral tegmental area, such as a slowly ramping membrane potential during constant current injections giving rise to delayed spiking and postinhibitory hypoexcitability (**Figure 1E**; [45, 46]). Moreover, like other dopaminergic populations in the midbrain, most DRN^{DA} neurons displayed rebound oscillations and the vast majority of cells expressed sag currents (**Figure 1E**, F). When comparing the electrophysiological properties of DRN^{DA} to DRN$^{5\text{-}HT}$ neurons, we observed numerous differences between these two cell types, but here we focus on the five most significant ones. While DRN$^{5\text{-}HT}$ neurons spike with short delays in response to current steps and maintain a relatively constant action potential (AP) amplitude, DRN^{DA} neurons display a longer delay to the first spike and the amplitude of subsequent action potentials drops (**Figure 1F-I**). Additionally, the action potentials of DRN$^{5\text{-}HT}$ neurons rise faster, while their afterhyperpolarization (AHP) is longer compared to DRN^{DA} neurons (**Figure 1H**, I). Lastly, the capacitance of DRN$^{5\text{-}HT}$ neurons is significantly larger than in DRN^{DA} neurons (**Figure 1I**).

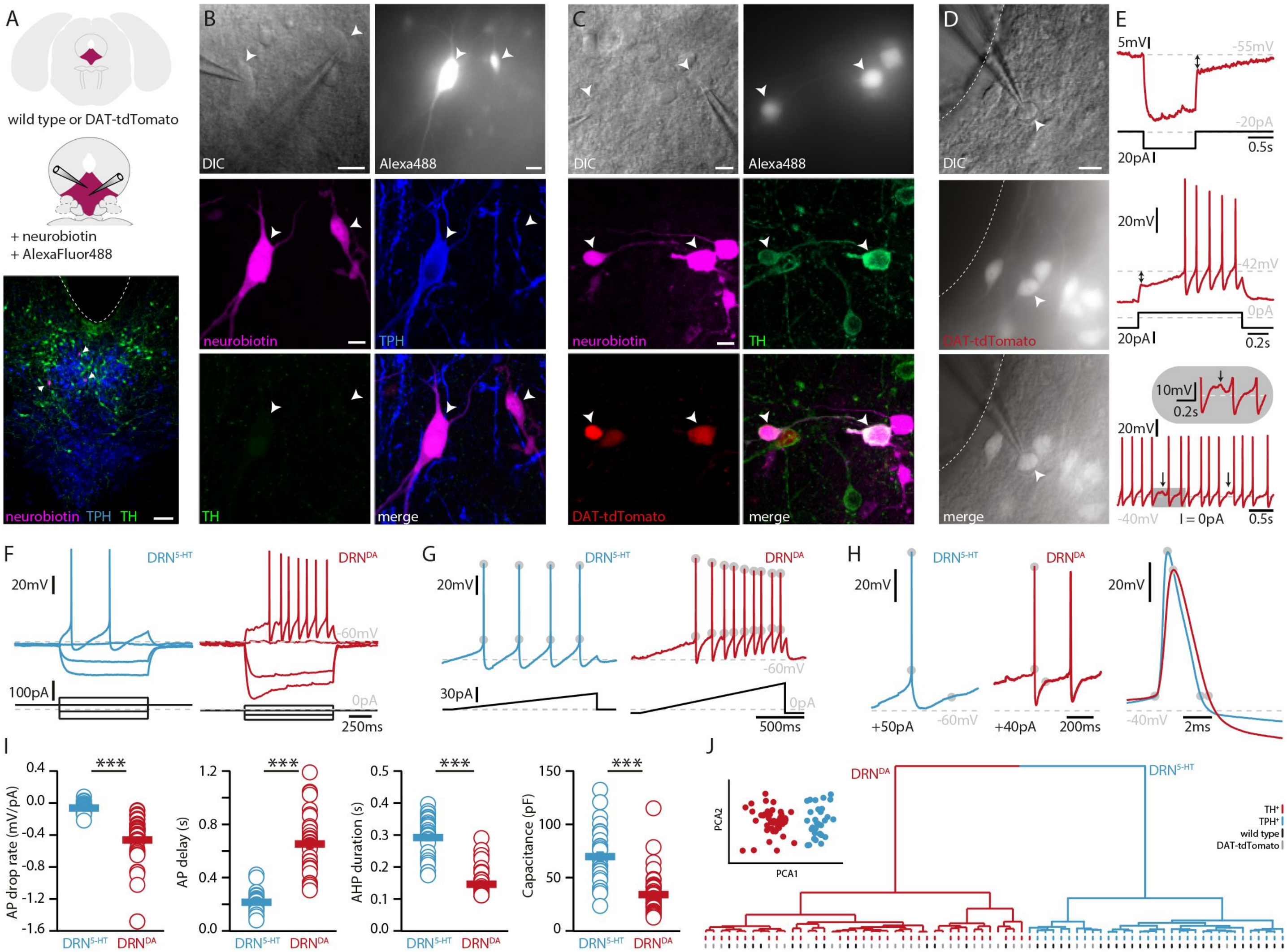

Figure 1.

DRN^{DA} and DRN$^{5\text{-}HT}$ are electrophysiologically distinct cell-types.

(A) Scheme of the location of the DRN (pink) in a coronal section (top) and at higher magnification together with two patch pipettes (center). Bottom: A representative slice stained post recording for TPH, TH and neurobiotin revealing serotonergic neurons (arrows). The ventricle is indicated with a dashed line. (B) Top: Differential interference contrast (DIC) microscopy image (left) of neurons that were filled with Alexa488 (right) and neurobiotin. Center, bottom: Staining of the same neurons revealing a TPH-positive (DRN$^{5\text{-}HT}$) neuron and a TPH- and TH-negative cell. (C) Top: DIC image of recorded neurons that were filled with Alexa488 and neurobiotin. Center, bottom: Staining of the same neurons revealing tdTomato- and TH-positive (DRN^{DA}) neurons. (D) Representative fluorescent (top), DIC (center) image and overlay (bottom) of a tdTomato-positive neuron in a DAT-tdTomato mouse. (E) Representative recordings depicting postinhibitory hypoexcitability, slowly ramping currents and rebound oscillations in DRN^{DA} neurons. (F) Representative voltage responses to current injections in a TPH-positive (DRN$^{5\text{-}HT}$) and TH-positive (DRN^{DA}) neuron. (G) Ramping current injections reveal action potential (AP) amplitude accommodation. Grey circles indicate the onset and peak of APs. (H) Amplitude and duration of the AP and action potential afterhyperpolarization (AHP) in a DRN$^{5\text{-}HT}$ and DRN^{DA} neurons. Grey circles indicate onset, peak, and end of the AP and AHP. (I) Quantification of electrophysiological properties distinguishing DRN$^{5\text{-}HT}$ from DRN^{DA} neurons (n= 28-43 cells per group, N = 9; Wilcoxon Rank Sum Test). (J) Principal component analysis (PCA) of five electrophysiological parameters (insert) and hierarchical cluster analysis based on PCA1 and PCA2 (Ward's method, Euclidean distance). Intrinsic properties were sufficient to separate TPH-positive cells (blue dash) from TH-positive (red dash) cells. Bottom dashes indicate wild type (black) and DAT-tdTomato (gray) mice. Data are shown as mean ± SEM, ***$p < 0.001$. Scale bars: A, 100 µm; B-D, 10 µm.

Next, we tested if DRN^{DA} neurons can be distinguished from DRN$^{5\text{-}HT}$ neurons based on these five electrophysiological parameters. To this end, we standardized the data and ran a principal component analysis (PCA) including all DRN$^{5\text{-}HT}$ neurons (i. e. all TPH-positive), all TH-positive neurons recorded in wild- type mice and all tdTomato-positive cells recorded in DAT-tdTomato mice (except for one outlier shown in Suppl. Figure 2F, G). Plotting the first two principal components (PC) showed two separate clusters (**Figure 1J**, insert). Unsupervised hierarchical cluster analysis based on PC1 and PC2 revealed the same two major clusters and potential subclusters (**Figure 1J**). Mapping the molecular identity of the cells onto the dendrogram revealed the separation of DRN$^{5\text{-}HT}$ and DRN^{DA} neurons, while there was no branching according to mouse line (wild type vs. DAT-tdTomato), further corroborating the validity of DAT-tdTomato mice as a marker for DRN^{DA} neurons. Overall, these data suggest that electrophysiological parameters themselves are sufficient to distinguish between DRN$^{5\text{-}HT}$ and DRN^{DA} neurons.

In addition to DRN^{DA} and DRN$^{5\text{-}HT}$ the DRN contains an unknown number of cell types and 47 out of 120 recorded neurons were neither TH-positive, nor TPH-positive and did not express tdTomato. To test whether DRN^{DA} can also be distinguished from those populations based on their electrophysiological profile, we ran a PCA on 20 standardized parameters and used the first three PCAs for unsupervised hierarchical clustering (Suppl. Figure 3). Our analysis suggests that there might be four major electrophysiological cell types in the DRN. In contrast to DRN^{DA} and DRN$^{5\text{-}HT}$ neurons, a large proportion of the remaining cells showed rebound spiking and biphasic AHPs, resembling the profiles of local interneurons in other brain areas (Suppl. Figure 3D, E). Interestingly, the clustering also indicated that three TH- and tdTomato-negative neurons belonged to the DRN^{DA} neurons and further analysis showed that they were indistinguishable from molecularly identified DRN^{DA} neurons (Suppl. Figure 3C, F). These findings indicate that clustering can be used to identify neurons that otherwise would have been excluded due to a lack of post-hoc staining data or genetic driver lines.

Overall, our data show that DRN^{DA} neurons constitute an electrophysiologically distinct class of neurons in the DRN expressing several hallmark properties, which are sufficient to identify them within the local DRN circuitry.

DRN^{DA} and DRN$^{5\text{-}HT}$ neurons have different morphological profiles

Next, we characterized the morphological profile of DRN$^{5\text{-}HT}$ and DRN^{DA} neurons. We focused on the analysis of somatic and dendritic properties since a complete reconstruction of the axonal arborization could not be retrieved from the slices. The analysis of the somatic properties showed that DRN$^{5\text{-}HT}$ neurons had larger cell bodies than DRN^{DA} neurons (**Figure 2A-E**), as measured in their area, perimeter, length, and width (**Figure 2B-E**). Cell bodies also differed in shape, with DRN^{DA} neurons having more circular somata than DRN$^{5\text{-}HT}$ neurons, as indicated by the circularity index (**Figure 2F**). Analyzing the dendritic properties, we found that DRN$^{5\text{-}HT}$ neurons had four to five primary dendrites, compared to only two to three in DRN^{DA} neurons (**Figure 2G**). Moreover, dendrites of DRN^{DA} neurons were frequently bipolar with the main primary dendrites starting from opposite extremes of the soma. Both populations had relatively few bifurcations (**Figure 2H**) but the DRN$^{5\text{-}HT}$ neurons had significantly more terminations (**Figure 2I**). The overall dendritic length did not differ between the DRN$^{5\text{-}HT}$ and DRN^{DA} neurons: both populations had a mix of short and long dendrites (**Figure 2L**). These data suggest that DRN$^{5\text{-}HT}$ neurons have denser dendritic arborization than DRN^{DA} neurons, mostly due to larger numbers of primary dendrites.

Altogether, our results show that DRN$^{5\text{-}HT}$ and DRN^{DA} neurons have distinct morphological properties. DRN$^{5\text{-}HT}$ neurons are mostly multipolar neurons, with a big and complex soma and multiple primary dendrites, while DRN^{DA} neurons have smaller and more circular cell bodies with bipolar dendrites.

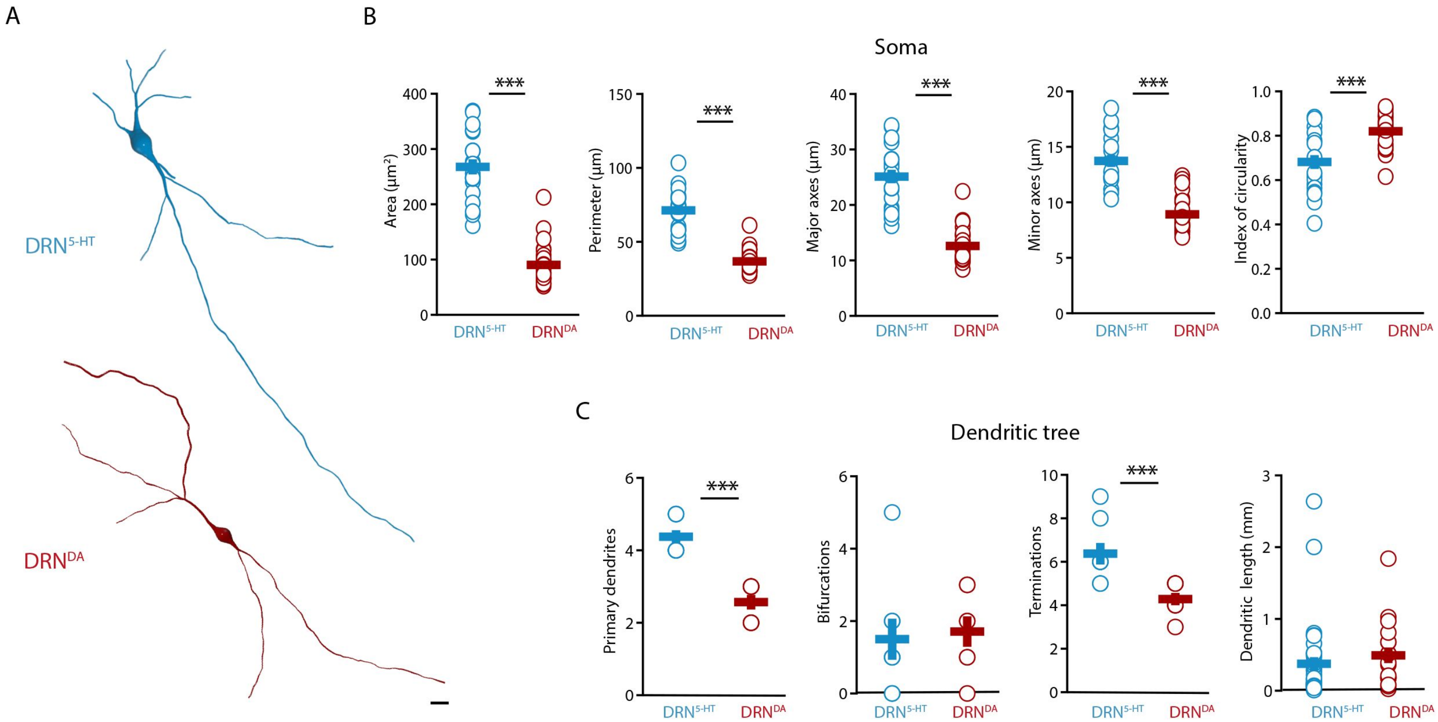

Figure 2.

DRN^{DA} and DRN$^{5\text{-}HT}$ have distinct morphological profiles.

(A) Top: representative digital reconstruction of a DRN$^{5\text{-}HT}$ from a wild-type mouse. Bottom: representative digital reconstruction of a DRN^{DA} from a wild-type mouse. Scale bar: 10 µm. (B-F) Morphological parameters describing the soma size and shape of DRN$^{5\text{-}HT}$ and DRN^{DA} neurons in Sham-lesion mice (DRN$^{5\text{-}HT}$: n = 20, N = 3; DRN^{DA}: n = 27, N = 3; unpaired t-test). (G-J) Morphological parameters describing the dendritic tree of DRN$^{5\text{-}HT}$ and DRN^{DA} neurons in Sham-lesion mice (DRN$^{5\text{-}HT}$: n = 8, N = 3; DRN^{DA}: n = 7, N = 3; unpaired t-test). Data are shown as mean ± SEM, ***$p < 0.001$.

DA and NA depletion distinctly affect the membrane properties of DRN$^{5\text{-}HT}$ neurons

To elucidate how DRN$^{5\text{-}HT}$ and DRN^{DA} neurons might be affected in PD, we characterized these populations in a mouse model of PD based on bilateral injection of the neurotoxin 6-OHDA in the dorsal striatum. This approach leads to a partial lesion of catecholamine neurons, reproducing an early stage of parkinsonism in which particularly non-motor symptoms are manifested [47]. In line with previous studies, we observed a 60-70% reduction of TH levels in the striatum (Suppl. Figure 4, [48]). Only mice meeting this criterion were included in the study. Immunostaining showed that the striatal 6-OHDA injection did not cause degeneration of DRN$^{5\text{-}HT}$ or DRN^{DA} neurons (Suppl. Figure 5).

Striatal injection of 6-OHDA has also been found to produce a partial loss of NA neurons in the LC [48]. In the present study, we determined the specific impact of NA dysfunction on the physiology of DRN$^{5\text{-}HT}$ and DRN^{DA} neurons by pre-treating a group of mice with desipramine (DMI), a selective inhibitor of noradrenaline reuptake, before injecting 6-OHDA (DMI+6-OHDA mice). We then assessed the intrinsic properties of DRN$^{5\text{-}HT}$ and DRN^{DA} neurons in Sham-lesion (Sham), 6-OHDA- and DMI+6-OHDA-treated mice. Whole-cell recordings obtained from DRN$^{5\text{-}HT}$ neurons in control mice revealed that 36% of DRN$^{5\text{-}HT}$ neurons were spontaneously active in slices and the proportion of intrinsically active neurons was similar in mice injected with 6-OHDA (control: n = 12/33 DRN$^{5\text{-}HT}$ neurons, 6-OHDA: n = 7/18 DRN$^{5\text{-}HT}$ neurons, **Figure 3A**). However, DRN$^{5\text{-}HT}$ neurons recorded in DMI+6-OHDA mice showed an increased excitability: in this condition 73% of DRN$^{5\text{-}HT}$ neurons were spontaneously active and DRN$^{5\text{-}HT}$ neurons displayed lower rheobase currents than in control and 6-OHDA mice (**Figure 3A**, B). Since the noradrenergic system is protected by DMI in these mice, these findings suggest that the 6-OHDA induced lesion of midbrain dopaminergic neurons evoked the increased firing of DRN$^{5\text{-}HT}$ neurons.

While the excitability of DRN$^{5\text{-}HT}$ neurons was not affected in 6-OHDA mice, we observed that their firing properties were profoundly altered: DRN$^{5\text{-}HT}$ neurons recorded in 6-OHDA mice displayed significantly smaller and shorter APs and AHPs compared to Sham mice (**Figure 3C-E**). Moreover, DRN$^{5\text{-}HT}$ neurons fired at higher frequencies and with shorter delays in response to current injections (**Figure 3F-H**). Finally, the membrane time constant of DRN$^{5\text{-}HT}$ neurons was shorter in 6-OHDA injected mice than in Sham mice (**Figure 3I**). Interestingly, these changes were not observed in 6-OHDA injected mice pre-treated with DMI, suggesting that the noradrenergic lesion – and not the dopaminergic lesion – underlies the changes in the firing properties of DRN$^{5\text{-}HT}$ neurons in 6-OHDA mice. Taken together, these results indicate that DRN$^{5\text{-}HT}$ neurons are affected in the 6-OHDA mouse model of PD. Specifically, lesions of the dopaminergic system increase the excitability of DRN$^{5\text{-}HT}$ neurons whereas lesions of the noradrenergic systems change the firing properties of DRN$^{5\text{-}HT}$ neurons.

6-OHDA lesion induced hypotrophy of DRN$^{5\text{-}HT}$ neurons

Morphological analysis revealed a reduced soma size of the DRN$^{5\text{-}HT}$ neurons in 6-OHDA mice, which was manifested in a decrease in area, perimeter, and major axes in comparison to control mice (**Figure 4A-C**). Moreover, the increase in the circularity of the 6-OHDA group indicated that the shape of the soma of DRN$^{5\text{-} HT}$ neurons was also altered by the lesion (**Figure 4C**). These modifications were not observed in DMI + 6- OHDA mice, suggesting that preserving the noradrenaline system protected the DRN$^{5\text{-}HT}$ neurons (Figure4A- C). Finally, the injection of 6-OHDA also caused a significant reduction in the number of primary dendrites of DRN$^{5\text{-}HT}$ neurons in the 6-OHDA group and a tendency to fewer termination points (**Figure 4D**). The number of bifurcations and the dendritic length were not affected by the lesion (**Figure 4D**). Globally, these

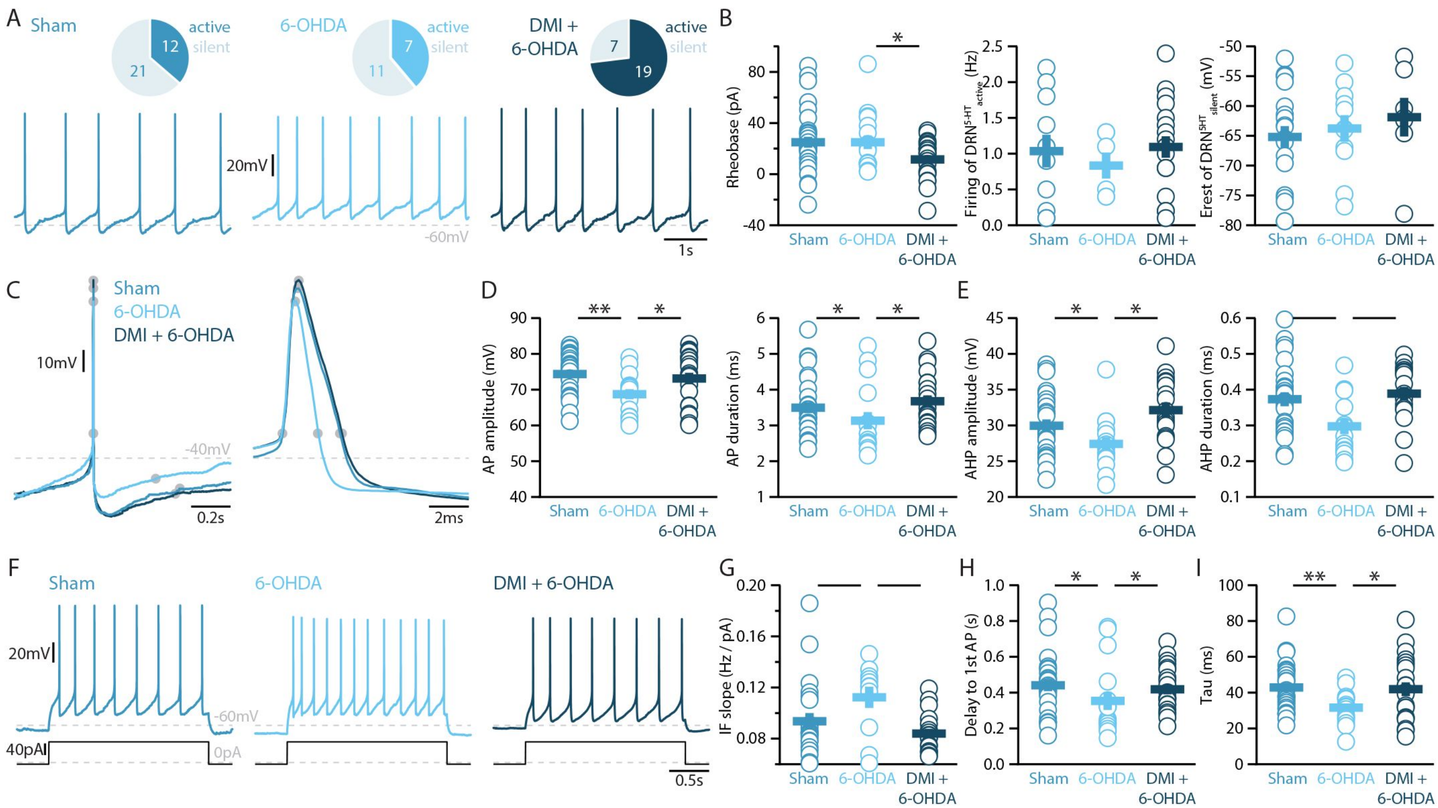

Figure 3.

Lesions targeting primarily nigrostriatal dopamine increase the excitability of DRN$^{5\text{-}HT}$ neurons whereas loss of NA affects their action potentials.

(A) Top: Pie charts showing the number of spontaneously active (dark) and silent (pale) DRN$^{5\text{-}HT}$ neurons in three conditions: Sham (left), 6-OHDA-injected mice (center) and 6-OHDA-injected mice pre- treated with desipramine (DMI + 6-OHDA, right). Bottom: Representative recordings of spontaneously active DRN$^{5\text{-}HT}$ neurons (I = 0 pA). (B) Quantification of the rheobase (left), the firing frequency of spontaneously active cells (center), and the resting membrane potential of silent DRN$^{5\text{-}HT}$ neurons (right). (C) Representative action potentials (APs) of DRN$^{5\text{-}HT}$ at low (left) and high (right) temporal. Grey circles indicate onset, offset, and peak of the APs as well as the end of the afterhyperpolarization (AHP). (D) Quantification of the amplitude (left) and duration (right) of the APs of DRN$^{5\text{-}HT}$ neurons. (E) Same as in (D) for the AHP. (F) Representative responses of DRN$^{5\text{-}HT}$ neurons to current steps (I = +75 pA). (G) Quantification of firing frequency / injected current. (H) Quantification of the delay to the first AP when injected with current eliciting 2Hz firing. (I) Quantification of the membrane time constant (tau) of DRN$^{5\text{-}HT}$ neurons (Sham: n = 22 - 32, N = 6 - 7; 6-OHDA: n = 11 - 16, N = 7; DMI + 6-OHDA: n = 18 – 21, N = 4; unpaired t-test). Data are shown as mean ± SEM, * $p < 0.05$, ** $p < 0.01$.

results suggest that the lesion produced by 6-OHDA induces a hypotrophic phenotype in DRN$^{5\text{-HT}}$ neurons characterized by a shrinkage of the soma and a decrease in dendritic branching and that these alterations are partly NA-dependent.

6-OHDA lesion affects the firing of DRN^{DA} neurons independent of NA loss

Finally, we assessed whether the striatal 6-OHDA lesion affects the physiology of DRN^{DA} neurons. Whole- cell patch-clamp recordings revealed that 58% of DRN^{DA} neurons are spontaneously active in slices of Sham- lesion mice (**Figure 5A**). In contrast, the proportion of intrinsically active neurons increased to 77% and 78% of DRN^{DA} neurons in 6-OHDA injected mice with and without pre-treatment with DMI, respectively. In line with this, silent DRN^{DA} neurons recorded in DMI + 6-OHDA mice were also found to have more depolarized resting membrane potentials and hence to be closer to action potential threshold (**Figure 5B**). Yet, active DRN^{DA} neurons tended to fire at lower frequencies in all 6-OHDA mice compared to control mice.

In stark contrast to DRN$^{5\text{-HT}}$ neurons, the APs and AHPs of DRN^{DA} neurons were not affected in any 6-OHDA mice (**Figure 5C-E**). In fact, we did not observe any change in the firing properties of DRN^{DA} neurons that were dependent on the protection of the NA system with DMI (**Figure 5**). Like DRN$^{5\text{-HT}}$ neurons, DRN^{DA} neurons displayed a reduction in spike latency in 6-OHDA mice, but the shortening of the delay was also present in DMI+6-OHDA mice. Together, these results suggest that the electrophysiological properties of DRN^{DA} neurons are affected in the 6-OHDA mouse model of PD and that these changes are primarily due to the lesion of the nigrostriatal dopaminergic pathway.

Striatal 6-OHDA injections increased soma complexity and dendritic branching in DRN^{DA} neurons

The morphological analysis of DRN^{DA} neurons revealed that the striatal 6-OHDA injection caused a significant reduction of the circularity from 0.82 (Sham) to 0.77 (6-OHDA) (**Figure 6A-C**), indicating increased complexity of the DRN^{DA} soma. We also observed that this effect was not present in the mice pre-treated with DMI (**Figure 6C**). The soma size was not affected by the lesion and no changes in the area, perimeter, length, and width were found (**Figure 6C**). However, dendritic branching, as shown by the increase in termination points and bifurcations in DRN^{DA} neurons was increased in 6-OHDA-injected mice, but not in mice treated with a combination of DMI and 6-OHDA (**Figure 6D**). The number of primary dendrites and the overall dendritic length were not affected by the 6-OHDA lesion (**Figure 6D**). Overall, these data suggest that impairment of noradrenergic function caused by the striatal 6-OHDA injection altered the soma shape and dendritic branching of DRN^{DA} neurons.

Discussion

In the present study we combine *ex vivo* whole-cell patch clamp recordings with morphological reconstructions and immunohistochemistry, to show that DRN^{DA} neurons have a distinct electrophysiological profile, which is sufficient to distinguish them from DRN$^{5\text{-HT}}$ neurons as well as other neuron classes in the DRN. Utilizing this approach, we also reveal that, in a 6-OHDA mouse model of PD, DRN$^{5\text{-HT}}$ neurons display distinct pathophysiological changes depending on the loss of dopamine and noradrenaline. Notably, degeneration of noradrenergic neurons affects not only the electrical properties of DRN$^{5\text{-HT}}$ neurons but also evokes hypotrophy of their cell bodies. In contrast, the loss of nigrostriatal dopamine mainly affects the electrophysiological properties of DRN^{DA} neurons while concomitant loss of noradrenaline alters their morphology.

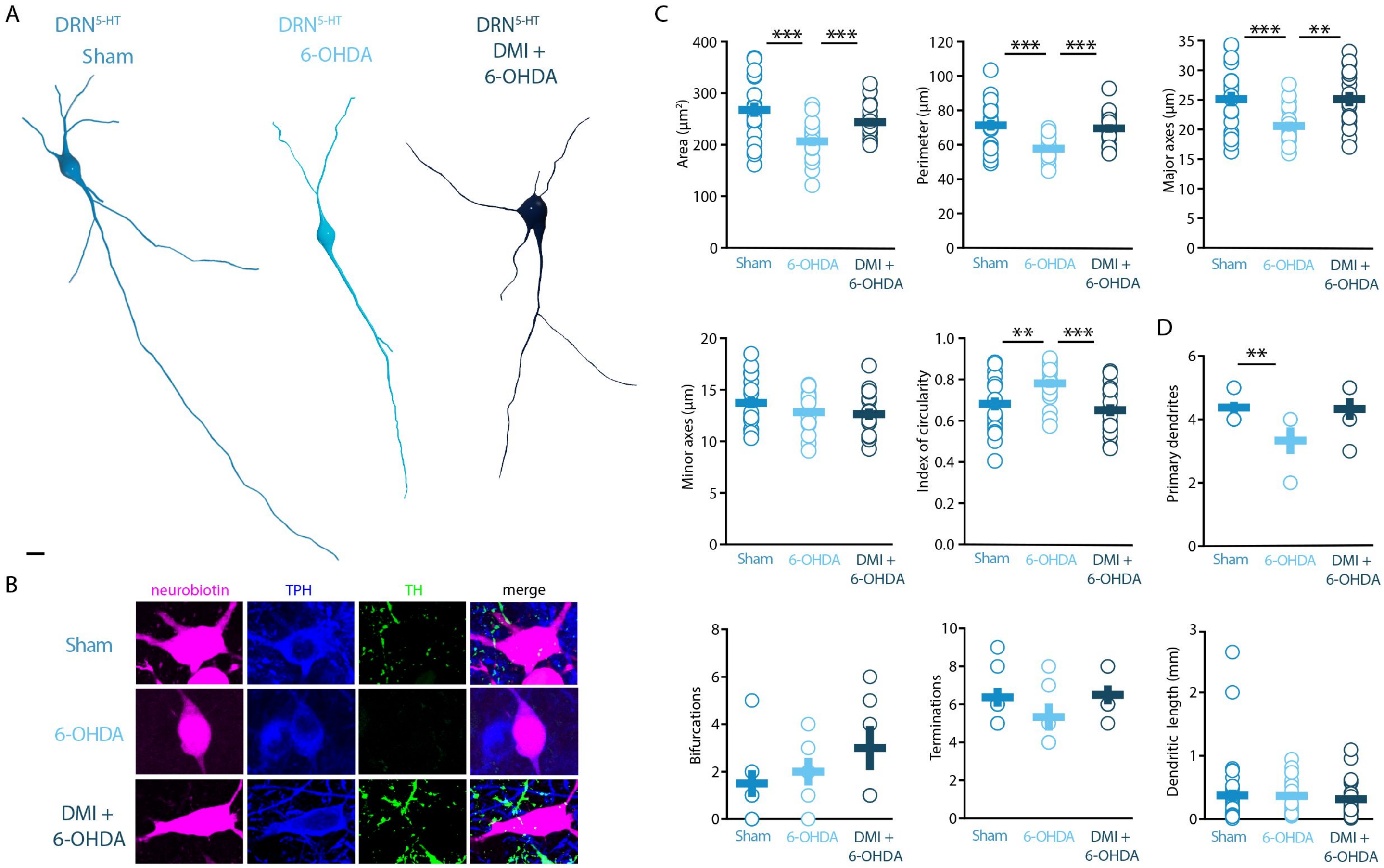

Figure 4.

Striatal injection of 6-OHDA induced a hypotrophic phenotype in the DRN$^{5\text{-}HT}$, which is prevented by pre- treatment with DMI.

(A) Representative digital reconstructions of a DRN$^{5\text{-}HT}$ neuron in three different conditions: Sham (left), 6-OHDA-injected mice (center) and 6-OHDA-injected mice pre-treated with desipramine (right). Scale bar: 10 µm (B) Representative confocal pictures of soma from DRN5-HT neurons in Sham (top), 6-OHDA-injected mice (center), and 6-OHDA-injected mice pre-treated with desipramine (bottom). Scale bar: 10 µm. (C-G) Morphological descriptors of the soma size and shape in DRN$^{5\text{-}HT}$ neurons (Sham: n = 20, N = 4; 6-OHDA: n = 19, N = 4; DMI + 6-OHDA: n = 17, N = 3; unpaired t-test).(H-K) Morphological descriptors of the dendritic tree in DRN$^{5\text{-}HT}$ neurons (Sham: n = 8, N = 3, 6-OHDA: n = 6, N = 3: DMI + 6-OHDA: n = 6, N = 2; unpaired t-test). Data are shown as mean ± SEM, ***p<0.001, **p<0.01. Scale bar: 10 µm.

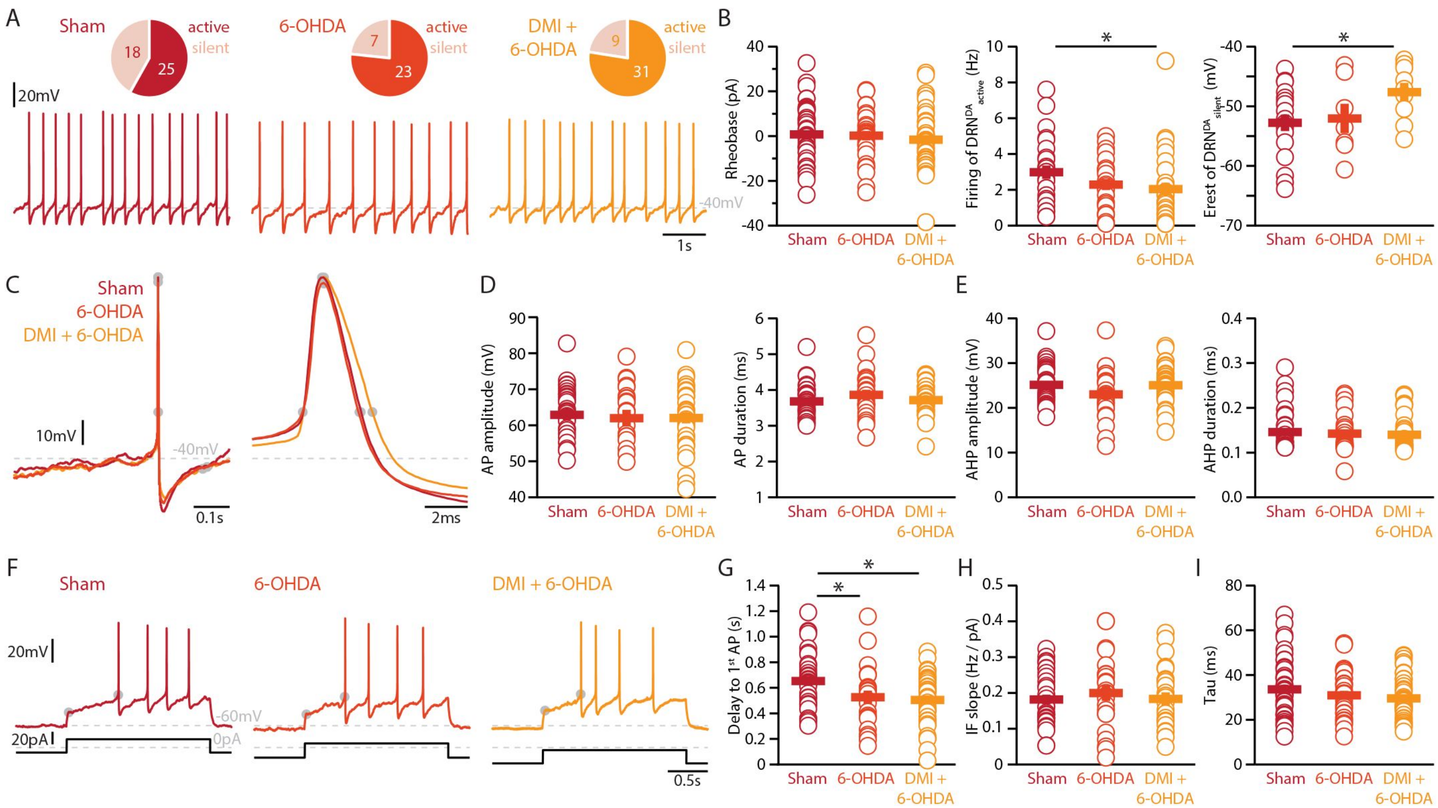

Figure 5.

Lesions targeting primarily SN dopamine depolarize DRN^{DA} neurons whereas concomitant loss of NA does not affect their action potentials.

(A) Top: Pie charts showing the proportion of spontaneously active (dark) and silent (pale) DRN^{DA} neurons in three conditions: Sham (left), 6OHDA-injected mice (center) and 6-OHDA-injected mice pre- treated desipramine (DMI, right). Bottom: Representative recordings of spontaneously active DRN^{DA} (I = 0pA). (B) Quantification of the rheobase (left), the firing frequency of spontaneously active (center), and the resting membrane potential of silent DRN^{DA} neurons (right). (C) Representative action potentials (APs) of DRN^{DA} at low (left) and high (right) temporal resolution. Grey circles indicate onset, offset, and peak of APs and the end of the afterhyperpolarization (AHP). (D) Quantification of the amplitude (left) and duration (right) of the APs of DRN^{DA} neurons. (E) Same as in (D) for the AHP. (F) Representative responses of DRN^{DA} neurons to current steps (I = 75pA). Grey circles indicate the delay to the first AP. (G-I) Quantification of firing frequency / injected current (G), the delay to the first AP when injected with current eliciting 2 Hz firing (H), and the membrane time constant (tau, I) of DRN^{DA} neurons recorded (Sham: n = 43, N = 8; 6-OHDA: n = 29, N = 6; DMI + 6-OHDA: n = 40, N = 6; unpaired t-test). Data are shown as mean ± SEM, * $p < 0.05$.

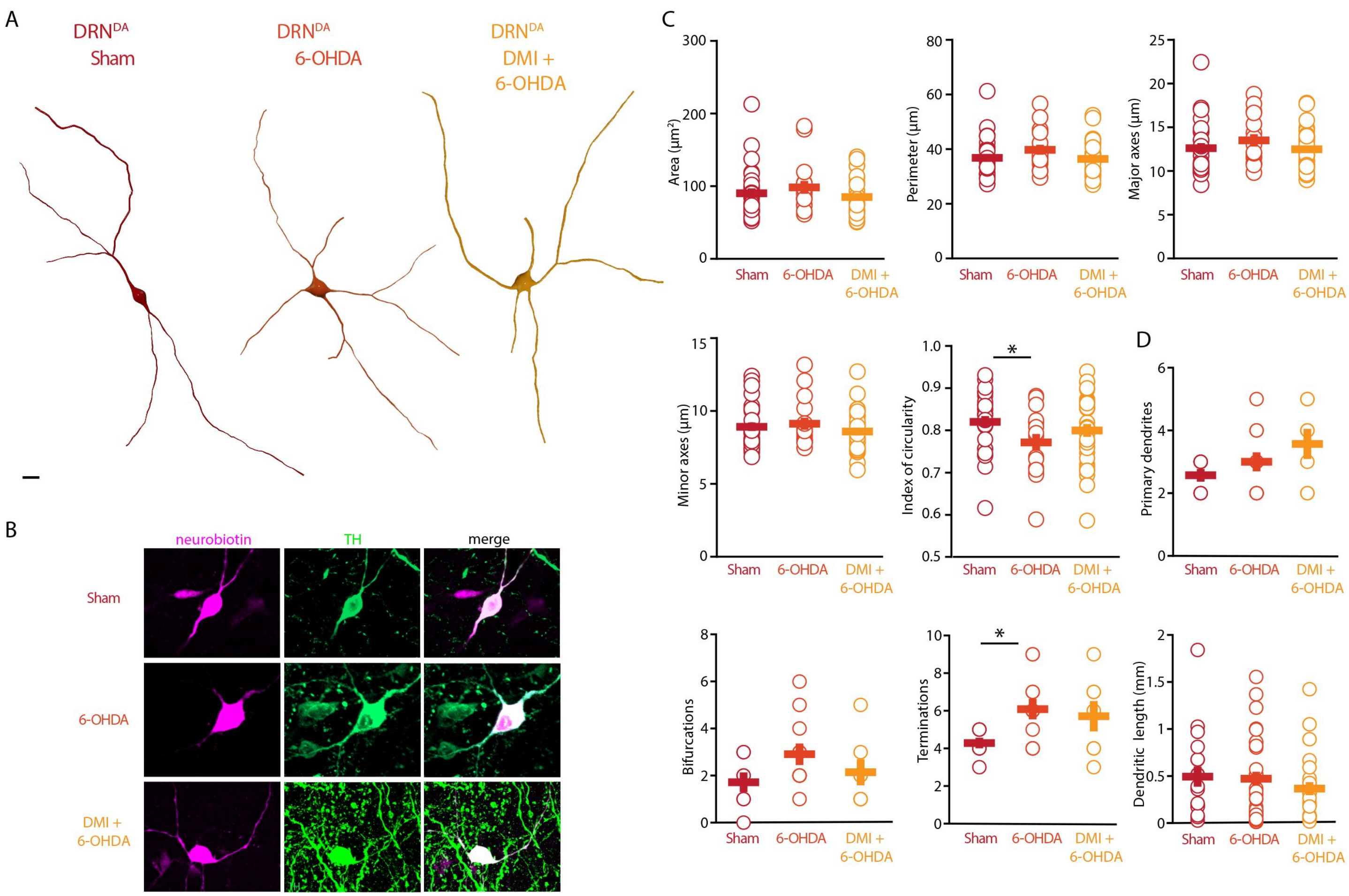

Figure 6

Striatal injection of 6-OHDA increased soma complexity and branching in DRN^{DA}.

(A) Representative digital reconstructions of a DRN^{DA} neuron in three different conditions: Sham (left), 6-OHDA-injected mice (center) and 6- OHDA-injected mice pre-treated with desipramine (right). (B) Representative confocal pictures of soma from DRN^{DA} neurons in Sham (top), 6-OHDA-injected mice (center), and 6-OHDA-injected mice pre-treated with DMI (bottom). (C- G) Morphological descriptors of the soma size and shape in DRN^{DA} neurons (Sham: n = 27, N = 7, 6-OHDA: n = 16, N = 4; DMI + 6-OHDA: n = 31, N = 5; unpaired t-test). (H-K) Morphological descriptors of the dendritic tree in DRN^{DA} neurons. (Sham: n = 7, N = 3; 6-OHDA: n = 11, N = 4; DMI + 6-OHDA: n = 7, N = 3; unpaired t-test). Data are shown as mean ± SEM, *p<0.05. Scale bar: 10 µm.

We used an extensive electrophysiological characterization protocol to quantify the differences between the DRN^{DA} and DRN$^{5\text{-HT}}$ populations. The electrophysiological properties agree with previous studies, such as the spontaneous firing pattern seen in DRN^{DA} neurons [33], and the slow AHP of DRN$^{5\text{-HT}}$ neurons [21]. Standard electrophysiological parameters were used to create a classification tool, which efficiently identifies DRN$^{5\text{-HT}}$ and DRN^{DA} cells, including dopamine neurons confirmed by TH staining and / or by fluorescent expression in DAT-tomato mice (**Figure 1**). Importantly, the DRN^{DA} neurons recorded from wild- type and DAT-tdTomato mice did not differ in their electrical properties, indicating that the transgene does not interfere with the membrane properties of this population.

We showed that DRN^{DA} neurons share electrophysiological properties with other dopaminergic populations in the midbrain [45, 46] but have electrophysiological profiles distinct from DRN$^{5\text{-HT}}$ neurons as well as other neuronal populations in the DRN. Most of the parameters extracted in our characterization rely on intracellular recordings of the membrane potential. However, some properties such as the spontaneous firing and action potential kinetics could be useful for *in vivo* characterization, even in extracellular recordings [49–51]. In addition to the DRN$^{5\text{-HT}}$ and DRN^{DA} neuronal populations, a large fraction of neurons displayed electrophysiological properties that were distinct from these two groups (Suppl. Figure 3), suggesting that there are other neuronal subtypes in the DRN network, such as previously reported GABAergic, glutamatergic, and peptidergic neurons [9, 52–57].

In line with previous studies, the majority of DRN$^{5\text{-HT}}$ neurons were large multipolar or fusiform neurons with four to five primary dendrites, very distinct from the DRN^{DA} neurons [58–60]. Very little is known about the morphology of the DRN^{DA} neurons, but previous studies identified small ovoid cells in the DRN which are likely to correspond to the DRN^{DA} cells [33, 61]. Out of 25 reconstructed DRN$^{5\text{-HT}}$ neurons, only one displayed dendritic spines. Previous studies in rats, described the presence of dendritic spines in most DRN$^{5\text{- HT}}$ neurons [62]. However, the study was performed in thicker slices and the dendritic spines were scarce in the primary and secondary dendrites, while they became dense in the distal dendrites, thus it is possible that in our study those dendrites were not present [62].

In the present study we assessed the impact on DRN cells of a striatal bilateral 6-OHDA lesion performed with or without DMI pre-treatment, which has been shown to protect the NA neurons from the 6-OHDA- induced degeneration [48, 63, 64]. We found that both DRN$^{5\text{-HT}}$ and DRN^{DA} populations were affected in a cell-type specific manner by the combined action of 6-OHDA on DA and NA, with DRN$^{5\text{-HT}}$ neurons being particularly sensitive to changes in the noradrenergic system. Loss of SNc dopaminergic neurons alone (6- OHDA + DMI) – which are known to target DRN$^{5\text{-HT}}$ neurons directly – increased the excitability and spontaneous activity in DRN$^{5\text{-HT}}$ neurons [53]. This is in line with previous *ex vivo* and *in vivo* studies showing that the DRN$^{5\text{-HT}}$ neurons display increased firing rates in rodents pre-treated with DMI and injected with 6-OHDA [21, 65] As hypothesized by Prinz et al. [21], the selective loss of midbrain DA may induce a homeostatic increase in the excitability of DRN$^{5\text{-HT}}$ neurons. Our data contrast with a previous *in vivo* study showing decreased firing activity in DRN$^{5\text{-HT}}$ neurons where injection of 6-OHDA was preceded by treatment with DMI and fluoxetine [22]. This dissimilarity may be related to species-specific (rat vs mouse) and technical (intracerebroventricular vs striatal injections, recordings performed at 10 days vs 3 weeks after the 6-OHDA injection). Importantly, in the same study identification of DRN$^{5\text{-HT}}$ neurons was not molecularly confirmed and the data may include other spontaneously active DRN neurons. In fact, our recordings show that there are non-serotonergic neurons in the DRN, which are spontaneously active and display a regular, slow firing frequency similar to DRN$^{5\text{-HT}}$ neurons, highlighting the importance of unequivocal identification of DRN cell types.

The present study shows that combined DA and NA lesioning affects DRN$^{5\text{-HT}}$ neurons more profoundly than selective loss of DA (**Figure 3**, **4**). In mice treated with 6-OHDA only, several electrophysiological and morphological properties were altered (**Figure 3**, **4**). The time constant and AHP of DRN$^{5\text{-HT}}$ neurons were shorter and the neurons responded with higher firing frequencies to current injections than in sham. This finding suggests that the pronounced AHP and long tau of these neurons may act as a "brake" limiting their maximum firing frequency in control conditions and that this brake is reduced when the NA system is lesioned. Future studies are needed to assess if DRN$^{5\text{-HT}}$ in fact fire at higher rates in vivo in mice treated with 6-OHDA. In contrast, such changes in DRN$^{5\text{-HT}}$ neurons were prevented when the NA system was protected by pre-treatment with DMI. These findings indicate an important role for NA as mediator of changes in the activity and properties of DRN$^{5\text{-HT}}$ neurons. The changes produced by the 6-OHDA lesion on the DRN^{DA} population were less pronounced than and different from those in DRN$^{5\text{-HT}}$ neurons. In terms of electrophysiological properties, the observed changes were primarily in the DA only lesion (6-OHDA + DMI), suggesting that unlike DRN$^{5\text{-HT}}$, DRN^{DA} neurons are affected by the loss of midbrain DA rather than the accompanying changes in NA (**Figure 5**). Conversely, the morphological changes produced by 6-OHDA in DRN^{DA} neurons appeared to require a concomitant DA and NA lesion, suggesting a stronger impact of the NA modulation (**Figure 6**).

Our results show that DRN neurons are affected by depletion of both DA and NA, thus raising the possibility that non-motor symptoms in PD are a result of the intricate organization of DA and NA neuromodulation as well as the interactions between the different DRN neuronal populations. Moreover, our results highlight the complex interplay between different neuromodulator systems [66], in this case NA, DA, and 5-HT in the DRN. The NA system is profoundly affected in PD [1, 8], but the precise pathophysiological processes resulting from loss of NA, and specifically the impact on DRN, are yet to be elucidated.

In conclusion, our study provides a quantitative description and classification scheme for two major neuronal populations in the dorsal raphe nucleus, DRN$^{5\text{-HT}}$ and DRN^{DA} neurons. We identified novel electrophysiological and morphological changes in these populations in response to DA and NA depletion in the basal ganglia. Considering the involvement of DRN and LC in the development of non-motor comorbidities, this study provides useful insights to understand better how these areas are affected in the parkinsonian condition. Moreover, our data pave the way for future experiments to characterize these subpopulations in terms of receptor expression and synaptic connectivity to shed light on their functional roles particularly with regard to the wide variety of non-motor symptoms observed in PD.

Methods

Experimental model details

All animal procedures were performed in accordance with the national guidelines and approved by the local ethics committee of Stockholm, Stockholms Norra djurförsöksetiska nämnd, under ethical permits to G. F. (N12148/17) and G. S. (N2020/2022). All mice (N=30) were group-housed under a 12 hr light / dark schedule and given ad libitum access to food and water. Wild-type mice ('C57BL/6J', #000664, the Jackson laboratory) and DAT-cre (Stock #006660 the Jackson laboratory) mice crossed with homozygous tdTomato reporter mice ('Ai9', stock #007909, the Jackson laboratory) were used.

6-OHDA model

Three-month old, male and female C57BL/6J or DAT-tdTomato were deeply anesthetized with Isoflurane and mounted on a stereotaxic frame (Stoelting Europe, Dublin, Ireland). To achieve a partial striatal lesion, each mouse received a bilateral injection of 1.25 µl of 6-hydroxydopamine hydrochloride (6-OHDA, (Sigma- Aldrich, 4µg/µl) or vehicle (0.9 % NaCl + Ascorbic Acid 0.02%) in the dorsolateral striatum, according to the following coordinates: anteroposterior +0.6 mm, mediolateral ± 2.2, dorsoventral -3.2 from Bregma, as previously described [47, 67]. One group of mice (referred to as DMI + 6-OHDA) was pre-treated with desipramine hydrochloride (DMI, Sigma-Aldrich, 25mg/kg i.p.) 30 minutes before the 6-OHDA infusion in order to protect the noradrenergic system [48].

Slice preparation and electrophysiology

Three weeks after the 6-OHDA injection, mice were deeply anaesthetized with isoflurane and decapitated. The brain was quickly removed and immersed in ice-cold cutting solution containing 205 mM sucrose, 10 mM glucose, 25 mM NaHCO3, 2.5 mM KCl, 1.25 mM NaH2PO4, 0.5 mM CaCl2 and 7.5 mM MgCl2. In all experiments, the brain was divided into two parts: the striatum was dissected from the anterior section for Western Blot and the posterior part was used to prepare coronal brain slices (250 um) with a Leica VT 1000S vibratome. Slices were incubated for 30-60 min at 34°C in a submerged chamber filled with artificial cerebrospinal fluid (ACSF) saturated with 95% oxygen and 5% carbon dioxide. ACSF was composed of 125 mM NaCl, 25 mM glucose, 25 mM NaHCO3, 2.5 mM KCl, 2 mM CaCl2, 1.25 mM NaH2PO4, and 1 mM MgCl2. Subsequently, slices were kept for at least 60 min at room temperature before recording.

Whole-cell patch clamp recordings were obtained in oxygenated ACSF at 35°C. Neurons were visualized using infrared differential interference contrast (IR-DIC) microscopy (Zeiss FS Axioskop, Oberkochen, Germany). DAT-tomato positive cells were identified by switching to epifluorescence using a mercury lamp (X-cite, 120Q, Lumen Dynamics). Up to three cells were patched simultaneously. Borosilicate glass pipettes (Hilgenberg) of 6 - 8 MOhm resistance were pulled with a Flaming / Brown micropipette puller P-1000 (Sutter Instruments). The intracellular solution contained 130 mM K-gluconate, 5 mM KCl, 10 mM HEPES, 4 mM Mg-ATP, 0.3 mM GTP, 10 mM Na2-phosphocreatine (pH 7.25, osmolarity 285 mOsm), 0.2% neurobiotin (Vector laboratories, CA) and Alexa-488 (75 µM) was added to the intracellular solution (Invitrogen). Spontaneous firing was recorded in cell-attached mode before break-in and all other whole- cell recordings were made in current-clamp mode. The intrinsic properties of the neurons were determined by a series of hyperpolarizing and depolarizing current steps and ramps, enabling the extraction of sub- and suprathreshold properties. Recordings were amplified using MultiClamp 700B amplifiers (Molecular Devices, CA, USA), filtered at 2kHz, digitized at 10-20kHz using ITC-18 (HEKA Elektronik, Instrutech, NY, USA), and acquired using custom-made routines running on IgorPro (Wavemetrics, OR, USA). Throughout all recordings pipette capacitance and access resistance were compensated for and data were discarded when access resistance increased beyond 30 MOhm. Liquid junction potential was not corrected for.

Immunofluorescence

Following the recordings, slices were fixated overnight at 4°C in 4% paraformaldehyde solution. Slices were then washed with PBS 1X. For the immunofluorescence, slices were treated with PBS 1X + Triton 0.3% and then incubated with a blocking solution of Normal Serum 10% and Bovine Serum Albumin 1% for 1h at room temperature. Afterward, slices were incubated overnight at 4°C with the following primary antibodies: rabbit anti-TH (Millipore, 1:1000), mouse anti-TPH (Sigma Aldrich, 1:600) and Streptavidin (Jackson Immunoresearch, 1:500). The following day, primary antibodies were washed out and slices were incubated with the appropriate fluorochrome-conjugated secondary antibodies.

For the immunostainings in the striatum and SNc and cell counting in DRN, mice were deeply anesthetized and transcardially perfused with PFA 4%. The brains were extracted and post-fixed in PFA 4% for 24 hours. 40 um coronal slices were prepared with a vibratome (Leica VT1000 S) and processed as described above.

Confocal Microscopy Analysis

The slices were imaged using Confocal (ZEISS LSM 800) at 10X and 40 X and z-stacks were retrieved. For cell identification, colocalization between neurobiotin and TH or TPH was evaluated.

Morphological analysis

For morphological analysis of dendrites, the confocal z-stacks were used in a semi-manual reconstruction using neuTube [68] and custom code, as previously described [69]. Soma morphology was analyzed by tracing manually the cell body profile, excluding dendritic trunks, in order to measure area (μm^2), perimeter, major and minor axis length (μm) and circularity values. Circularity, calculated as the ratio between the squared perimeter and the area (i.e. perimeter^2/4π area), can be a value between 0 and 1 (1 for circular shapes and values < 1 for more complex shapes.

The morphological analysis was performed on the neurobiotin stacks.

Western Blot

The striata were sonicated in 1% sodium dodecyl sulfate and boiled for 10 min. Equal amounts of protein (25 μg) for each sample were loaded onto 10% polyacrylamide gels and separated by electrophoresis and transferred overnight to nitrocellulose membranes (Thermo Fisher, Stockholm, Sweden). The membranes were immunoblotted with primary antibodies against actin (1:30,000, Sigma Aldrich, Stockholm, Sweden) and TH (1:2000, Millipore, Darmstadt, Germany). Detection was based on fluorescent secondary antibody binding (IR Dye 800CW and 680RD, Li-Cor, Lincoln, NE, USA) and quantified using a Li-Cor Odyssey infrared fluorescent detection system (Li-Cor, Lincoln, NE, USA). The TH protein levels were normalized for the amount of the corresponding actin detected in the sample and then expressed as a percentage of the control (Sham-lesion).

Statistical Analysis

Statistical analysis of morphology data was performed using GraphPad Prism 9.2.0. Data are reported as average as ± SEM and analyzed by unpaired t-test. N indicates the number of mice, while n indicates the number of cells. Significance was set at $p<0.05$.

Acknowledgements

We thank Elin Dahlberg for technical assistance and Kristoffer Tenebro Berglund for taking care of the mice. We also thank the members of the Silberberg and Fisone labs and the AND-PD consortium members for comments and discussions.

Funding

This work was supported by a Wallenberg Fellowship from the Knut & Alice Wallenberg Foundation (KAW 2017.0273), the Swedish Brain Foundation (Hjärnfonden, FO2021-0333), and the Swedish Medical Research Council (VR-M, 2019-0 1254) to GS, and a Swedish Research Council International Postdoc Grant (2020-06365) to YJ. RT, RA, GF, and GS are supported by an EU grant (H2020 848002 AND-PD).

Author contributions

Conceptualization: LB, YJ, RT, RM, GF, GS.

Investigation: YJ (electrophysiology, clustering analysis), LB (6-OHDA lesions, morphological analysis, cell counting, western blot).

Visualization: YJ (electrophysiological and clustering results), LB (6-OHDA model, morphological reconstructions).

Supervision: GS and GF. Writing—original draft: YJ and LB. Writing—review & editing: LB, YJ, GF, GS.

Competing Interests

The authors declare that they have no competing interests.

Data availability

All raw data is available upon reasonable request.

References

1. Braak H., et al. (2003) **Staging of brain pathology related to sporadic Parkinson's disease** *Neurobiol Aging* **24**:197–211

2. Jankovic J (2008) **Parkinson's disease: clinical features and diagnosis** *J Neurol Neurosurg Psychiatry* **79**:368–76

3. Swick T.J (2012) **Parkinson's disease and sleep/wake disturbances** *Parkinsons Dis* **2012**

4. Chaudhuri K.R., Schapira A.H. (2009) **Non-motor symptoms of Parkinson's disease: dopaminergic pathophysiology and treatment** *Lancet Neurol* **8**:464–74

5. Halliday G.M., et al. (1990) **Loss of brainstem serotonin- and substance P-containing neurons in Parkinson's disease** *Brain Res* **510**:104–7

6. Halliday G.M., et al. (1990) **Neuropathology of immunohistochemically identified brainstem neurons in Parkinson's disease** *Ann Neurol* **27**:373–85

7. Gesi M., et al. (2000) **The role of the locus coeruleus in the development of Parkinson's disease** *Neurosci Biobehav Rev* **24**:655–68

8. Zarow C., et al. (2003) **Neuronal loss is greater in the locus coeruleus than nucleus basalis and substantia nigra in Alzheimer and Parkinson diseases** *Arch Neurol* **60**:337–41

9. Huang K.W., et al. (2019) **Molecular and anatomical organization of the dorsal raphe nucleus** *Elife* **8**

10. Maillet A., et al. (2021) **Serotonergic and Dopaminergic Lesions Underlying Parkinsonian Neuropsychiatric Signs** *Mov Disord*

11. Jørgensen L.M., et al. (2021) **Parkinson patients have a presynaptic serotonergic deficit: A dynamic deep brain stimulation PET study** *J Cereb Blood Flow Metab* **41**:1954–1963

12. Cheshire P., et al. (2015) **Serotonergic markers in Parkinson's disease and levodopa-induced dyskinesias** *Mov Disord* **30**:796–804

13. Pifl C., Schingnitz G., Hornykiewicz O. (1991) **Effect of 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine on the regional distribution of brain monoamines in the rhesus monkey** *Neuroscience* **44**:591–605

14. Rylander D., et al. (2010) **Maladaptive plasticity of serotonin axon terminals in levodopa-induced dyskinesia** *Ann Neurol* **68**:619–28

15. Nayyar T., et al. (2009) **Cortical serotonin and norepinephrine denervation in parkinsonism: preferential loss of the beaded serotonin innervation** *Eur J Neurosci* **30**:207–16

16. Taylor T.N., et al. (2009) **Nonmotor symptoms of Parkinson's disease revealed in an animal model with reduced monoamine storage capacity** *J Neurosci* **29**:8103–13

17. Karstaedt P.J., et al. (1994) **Unilateral destruction of dopamine pathways increases ipsilateral striatal serotonin turnover in rats** *Exp Neurol* **126**:25–30

18. Wilson H., et al. (2018) **Serotonergic dysregulation is linked to sleep problems in Parkinson's disease** *Neuroimage Clin* **18**:630–637

19. Politis M., et al. (2010) **Depressive symptoms in PD correlate with higher 5-HTT binding in raphe and limbic structures** *Neurology* **75**:1920–7

20. Kaya A.H., et al. (2008) **Increased electrical and metabolic activity in the dorsal raphe nucleus of Parkinsonian rats** *Brain Res* **1221**:93–7

21. Prinz A., et al. (2013) **Increased excitability in serotonin neurons in the dorsal raphe nucleus in the 6-OHDA mouse model of Parkinson's disease** *Exp Neurol* **248**:236–45

22. Guiard B.P., et al. (2008) **Functional interactions between dopamine, serotonin and norepinephrine neurons: an in-vivo electrophysiological study in rats with monoaminergic lesions** *Int J Neuropsychopharmacol* **11**:625–39

23. Wenk G.L., et al. (1994) **Behavioral, biochemical, histological, and electrophysiological effects of 192 IgG-saporin injections into the basal forebrain of rats** *J Neurosci* **14**:5986–95

24. Lu J., Jhou T.C., Saper C.B. (2006) **Identification of wake-active dopaminergic neurons in the ventral periaqueductal gray matter** *J Neurosci* **26**:193–202

25. Dzirasa K., et al. (2006) **Dopaminergic control of sleep-wake states** *J Neurosci* **26**:10577–89

26. Cho J.R., et al. (2017) **Dorsal Raphe Dopamine Neurons Modulate Arousal and Promote Wakefulness by Salient Stimuli** *Neuron* **94**:1205–1219

27. Lin R., et al. (2020) **The Raphe Dopamine System Controls the Expression of Incentive Memory** *Neuron* **106**:498–514

28. Matthews G.A., et al. (2016) **Dorsal Raphe Dopamine Neurons Represent the Experience of Social Isolation** *Cell* **164**:617–31

29. Taylor N.E., et al. (2019) **The Role of Glutamatergic and Dopaminergic Neurons in the Periaqueductal Gray/Dorsal Raphe: Separating Analgesia and Anxiety** *eNeuro* **6**

30. Li C., et al. (2016) **Mu Opioid Receptor Modulation of Dopamine Neurons in the Periaqueductal Gray/Dorsal Raphe: A Role in Regulation of Pain** *Neuropsychopharmacology* **41**:2122–32

31. Flores J.A., et al. (2004) **Opiate anti-nociception is attenuated following lesion of large dopamine neurons of the periaqueductal grey: critical role for D1 (not D2) dopamine receptors** *Pain* **110**:205–14

32. Flores J.A., et al. (2006) **Role for dopamine neurons of the rostral linear nucleus and periaqueductal gray in the rewarding and sensitizing properties of heroin** *Neuropsychopharmacology* **31**:1475–88

33. Dougalis A.G., et al. (2012) **Functional properties of dopamine neurons and co-expression of vasoactive intestinal polypeptide in the dorsal raphe nucleus and ventro-lateral periaqueductal grey** *Eur J Neurosci* **36**:3322–3332

34. Pinto Cardozo, et al D.F. (2019) **Characterization of transgenic mouse models targeting neuromodulatory systems reveals organizational principles of the dorsal raphe** *Nat Commun* **10**

35. Fu W., et al. (2010) **Chemical neuroanatomy of the dorsal raphe nucleus and adjacent structures of the mouse brain** *J Comp Neurol* **518**:3464–94

36. Okaty B.W., et al. (2015) **Multi-Scale Molecular Deconstruction of the Serotonin Neuron System** *Neuron* **88**:774–91

37. Lindeberg J., et al. (2004) **Transgenic expression of Cre recombinase from the tyrosine hydroxylase locus** *Genesis* **40**:67–73

38. Lammel S., et al. (2015) **Diversity of transgenic mouse models for selective targeting of midbrain dopamine neurons** *Neuron* **85**:429–38

39. Carter M.E., et al. (2010) **Tuning arousal with optogenetic modulation of locus coeruleus neurons** *Nat Neurosci* **13**:1526–33

40. Pudovkina O.L., Cremers T.I., Westerink B.H. (2003) **Regulation of the release of serotonin in the dorsal raphe nucleus by alpha1 and alpha2 adrenoceptors** *Synapse* **50**:77–82

41. Trulson M.E., Crisp T. (1984) **Role of norepinephrine in regulating the activity of serotonin-containing dorsal raphe neurons** *Life Sci* **35**:511–5

42. Haddjeri N., de Montigny C., Blier P. (1997) **Modulation of the firing activity of noradrenergic neurones in the rat locus coeruleus by the 5-hydroxtryptamine system** *Br J Pharmacol* **120**:865–75

43. Singewald N., et al. (1998) **Influence of excitatory amino acids on basal and sensory stimuli- induced release of 5-HT in the locus coeruleus** *Br J Pharmacol* **123**:746–52

44. Aston-Jones G., et al. (1991) **Serotonin selectively attenuates glutamate-evoked activation of noradrenergic locus coeruleus neurons** *J Neurosci* **11**:760–9

45. Lammel S., et al. (2008) **Unique properties of mesoprefrontal neurons within a dual mesocorticolimbic dopamine system** *Neuron* **57**:760–73

46. Neuhoff H., et al. (2002) **I(h) channels contribute to the different functional properties of identified dopaminergic subpopulations in the midbrain** *J Neurosci* **22**:1290–302

47. Bonito-Oliva A., Masini D., Fisone G. (2014) **A mouse model of non-motor symptoms in Parkinson's disease: focus on pharmacological interventions targeting affective dysfunctions** *Front Behav Neurosci* **8**

48. Bonito-Oliva A., et al. (2014) **Cognitive impairment and dentate gyrus synaptic dysfunction in experimental parkinsonism** *Biol Psychiatry* **75**:701–10

49. Strickland J.A., McDannald M.A. (2022) **Brainstem networks construct threat probability and prediction error from neuronal building blocks** *Nat Commun* **13**

50. Hajós M., et al. (2007) **Neurochemical identification of stereotypic burst-firing neurons in the rat dorsal raphe nucleus using juxtacellular labelling methods** *Eur J Neurosci* **25**:119–26

51. Schweimer J.V., et al. (2011) **Spike-timing relationship of neurochemically-identified dorsal raphe neurons during cortical slow oscillations** *Neuroscience* **196**:115–23

52. Dahlstroem A., Fuxe K. (1964) **EVIDENCE FOR THE EXISTENCE OF MONOAMINE-CONTAINING NEURONS IN THE CENTRAL NERVOUS SYSTEM. I. DEMONSTRATION OF MONOAMINES IN THE CELL BODIES OF BRAIN STEM NEURONS** *Acta Physiol Scand Suppl* **232**:1–55

53. Pollak Dorocic I., et al. (2014) **A whole-brain atlas of inputs to serotonergic neurons of the dorsal and median raphe nuclei** *Neuron* **83**:663–78

54. Weissbourd B., et al. (2014) **Presynaptic partners of dorsal raphe serotonergic and GABAergic neurons** *Neuron* **83**:645–62

55. Gocho Y., et al. (2013) **Electrophysiological and pharmacological properties of GABAergic cells in the dorsal raphe nucleus** *J Physiol Sci* **63**:147–54

56. Xu Z., et al. (2021) **Whole-brain connectivity atlas of glutamatergic and GABAergic neurons in the mouse dorsal and median raphe nuclei** *Elife* **10**

57. Bowker R.M., et al. (1983) **Descending serotonergic, peptidergic and cholinergic pathways from the raphe nuclei: a multiple transmitter complex** *Brain Res* **288**:33–48

58. Rivera H.M., Bethea C.L. (2013) **Ovarian steroids increase PSD-95 expression and dendritic spines in the dorsal raphe of ovariectomized macaques** *Synapse* **67**:897–908

59. Park M.R (1987) **Intracellular horseradish peroxidase labeling of rapidly firing dorsal raphe projection neurons** *Brain Res* **402**:117–30

60. Calizo L.H., et al. (2011) **Raphe serotonin neurons are not homogenous: electrophysiological, morphological and neurochemical evidence** *Neuropharmacology* **61**:524–43

61. Diaz-Cintra S., et al. (1981) **Nucleus raphe dorsalis: a morphometric Golgi study in rats of three age groups** *Brain Res* **207**:1–16

62. Li Y.Q., et al. (2001) **Morphological features and electrophysiological properties of serotonergic and non-serotonergic projection neurons in the dorsal raphe nucleus. An intracellular recording and labeling study in rat brain slices** *Brain Res* **900**:110–8

63. Kamińska K., et al. (2017) **Depressive-like neurochemical and behavioral markers of Parkinson's disease after 6-OHDA administered unilaterally to the rat medial forebrain bundle** *Pharmacol Rep* **69**:985–994

64. Fulceri F., et al. (2006) **Nigrostriatal damage with 6-OHDA: validation of routinely applied procedures** *Ann N Y Acad Sci* **1074**:344–8

65. Wang S., et al. (2009) **Unilateral lesion of the nigrostriatal pathway induces an increase of neuronal firing of the midbrain raphe nuclei 5-HT neurons and a decrease of their response to 5-HT(1A) receptor stimulation in the rat** *Neuroscience* **159**:850–61

66. Mei J., Muller E., Ramaswamy S. (2022) **Informing deep neural networks by multiscale principles of neuromodulatory systems** *Trends Neurosci* **45**:237–250

67. Masini D., et al. (2021) **A Guide to the Generation of a 6-Hydroxydopamine Mouse Model of Parkinson's Disease for the Study of Non-Motor Symptoms** *Biomedicines* **9**

68. Feng L., Zhao T., Kim J. (2015) **neuTube 1.0: A New Design for Efficient Neuron Reconstruction Software Based on the SWC Format** *eNeuro* **2**

69. Hjorth J.J.J., et al. (2020) **The microcircuits of striatum in silico** *Proc Natl Acad Sci U S A* **117**:9554–9565

Author information

Laura Boi
Department of Neuroscience, Karolinska Institutet, Stockholm 17177, Sweden

Yvonne Johansson
Sainsbury Wellcome Centre for Neural Circuits and Behaviour, University College London, London, UK

Raffaella Tonini
Neuromodulation of Cortical and Subcortical Circuits Laboratory, Istituto Italiano di Tecnologia, Genova, Italy

Rosario Moratalla
Cajal Institute, Spanish National Research Council (CSIC), Madrid, Spain, CIBERNED, Instituto de Salud Carlos III, Madrid, Spain

Gilberto Fisone
Department of Neuroscience, Karolinska Institutet, Stockholm 17177, Sweden
For correspondence: gilad.silberberg@ki.se

Gilad Silberberg
Department of Neuroscience, Karolinska Institutet, Stockholm 17177, Sweden
For correspondence: gilad.silberberg@ki.se
ORCID iD: 0000-0001-9964-505X

Editors

Reviewing Editor
Jun Ding
Stanford University, United States of America

Senior Editor
John Huguenard
Stanford University School of Medicine, United States of America

Reviewer #1 (Public Review):

Summary:
People with Parkinson's disease often experience a variety of nonmotor symptoms, the biological bases of which remain poorly understood. Johansson et al began to study the

potential roles of the dorsal raphe nucleus (DRN) degeneration in the pathophysiology of neuropsychiatric symptoms in PD.

Strengths:
Johansson et al validated a transgenic reporter mouse line that can reliably label dopaminergic neurons in the DRN. This brain region shows severe neurodegeneration and has been proposed to contribute to the manifestation of neuropsychiatric symptoms in PD. Using this mouse line (and others), Johansson and colleagues characterized electrophysiological and morphological phenotypes of dopaminergic and serotoninergic neurons in the raphe nucleus. This study involved very careful topographical registration of recorded neurons to brain slices for post hoc immunohistochemical validation of cell identification, making it an elegant and thorough piece of work.

In relevance to PD pathophysiology, the authors evaluated the physiological and morphological changes of DRN serotoninergic and dopaminergic neurons after a partial loss of nigrostriatal dopamine neurons, which serves as a mouse model of early parkinsonian pathology. Importantly, the authors identified a series of physiological and morphological changes of subtypes of DRN neurons that depend on nigral dopaminergic neurodegeneration, LC adrenergic neurodegeneration, or both.

Overall, the study was well-designed, and the data were well-presented in this well-written manuscript.

Weaknesses:
Caveats that should be mentioned include:

1. While desipramine experiments provide clues about the potential role of adrenaline loss in electrophysiological and morphological changes in the Figs. 3-5, a complementary set of experiments is needed to confirm these findings. For example, how might selective LC adrenergic neurodegeneration affect cellular physiology and morphology in the DRN? Can the observed phenotypes in Figs 3-4 be rescued by adrenergic receptor agonists?

2. It should be kept in mind that the key experiments of this study were conducted using mouse models of parkinsonism. Thus, these models cannot recapitulate the complexity of PD pathology and circuit dysfunction.

Reviewer #2 (Public Review):

In this paper, Boi et al. thoroughly classified the electrophysiological and morphological characteristics of serotonergic and dopaminergic neurons in the DRN and examined the alterations of these neurons in the 6-OHDA-induced mouse PD model. Using whole-cell patch clamp recording, they found that 5-HT and dopamine (DA) neurons in the DRN are electrophysiologically well-distinguished from each other. In addition, they characterized distinct morphological features of 5-HT and DA neurons in the DRN. Notably, these specific features of 5-HT and DA neurons in the DRN exhibited different changes in the 6-OHDA-induced PD model. Then the authors utilized desipramine (DMI) to separate the effects of nigrostriatal DA depletion and noradrenalin (NA) depletion which are induced by 6-OHDA. Interestingly, protection from NA depletion by DMI pretreatment reversed the changes in 5-HT neurons, while having a minor impact on the changes in DA neurons in the DRN. These data indicate that the role of NA lesion in the altered properties of DRN 5-HT neurons by 6-OHDA is more critical than the one of DA lesion.

Overall, this study provides foundational data on the 5-HT and DA neurons in the DRN and their potential involvement in PD symptoms. Given the defects of the DRN in PD, this paper may offer insights into the cellular mechanisms that may underlie non-motor symptoms associated with PD. Despite the importance of the primary claim proposed by the authors, however, several weaknesses undermine the significance of the data.

Reviewer #3 (Public Review):

Summary:
Using ex vivo electrophysiology and morphological analysis, Boi et al. investigate the electrophysiological and morphological properties of serotonergic and dopaminergic subpopulations in the dorsal raphe nucleus (DRN). They performed labor-intensive and rigorous electrophysiology with posthoc immunohistochemistry and neuronal reconstruction to delineate the two major cell classes in the DRN: DRN-DA and DRN-5HT, named according to their primary neurotransmitter machinery. They find that the dopaminergic (DRN-DA) and serotonergic (DRN-5HT) neurons are electrophysiologically and morphologically distinct, and are altered following striatal injection of the toxin 6-OHDA. However, these alterations were largely prevented in DRN-5HT neurons by pre-treatment with desipramine. These findings suggest an important interplay between catecholaminergic systems in healthy and parkinsonian conditions, as well as a relationship between neuronal structure and function.

Strengths:
A large, well-validated dataset that will be a resource for others.
Complementary electrophysiological and anatomical characterizations.
Conclusions are justified by the data.
Relevant for basic scientists interested in DRN cell types and physiology.
Relevant for those interested in serotonin and/or DRN neurons in Parkinson's Disease.

Weaknesses:
Given the scope of the author's questions and hypotheses, I did not identify any major weaknesses.